Mid-Term test

Vithun P
10th 'C'
Roll: 65918.

* PART-A.

1. (c) Ampere.

2. (c) Between P and F.
Object between pole and focus → Virtual, erect, magnified image.

3. -

4. (b) 0.5m.

5. (c) All go off.

* PART-B.

6. $R_1 = 4\Omega$, $R_2 = 6\Omega$, $R_3 = 12\Omega$, $V = 12V$.

(a) $\frac{1}{R_P} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12} = \frac{6}{12} = \frac{1}{2} \Rightarrow R_P = 2\Omega$.

(b) $I = \frac{V}{R} = \frac{12}{2} = 6A$.

(c) $I_4 = V/R_1 = 12/4 = 3A$.

Q7. $u = -30cm$, $f = -10cm$.

(a) Mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$.

$\frac{1}{-10} - \frac{1}{(-30)} = -\frac{3}{30} + \frac{1}{30} = \frac{-2}{30} \Rightarrow v = -15cm$

(b) $m = -v/u = (-15)/(-30) = -15/30 = -0.5$.

(c) -

Q8. -

PART - C.

Q10. Appliances:

4 bulbs $= 4 \times 60 = 240W$

$\Rightarrow$ Refrigerator $= 150W$, TV $= 120W$, fan $= 75W$

(a) Total power $= 240 + 150 + 120 + 75 = 585 W$

(b) $I = P/V = 585/220 = 2.7A$ (Approx)

(c) Fuse rating $= 5A$, Current drawn $= 2.7A$

$2.7A < 5A \rightarrow$ Fuse will NOT blow.

(d) Safe fuse = Next standard rating above 2.7A.
$= 3A$ fuse

(e) Energy $= P \times t = 585 \times 8 = 4680 Wh = 4.68 Kwh$

$= 4.7 Kwh$.